

Selected robustness metrics

Continuous values: mean +/- sample SD; control roughness is mean per transition (lower is smoother); roughness/effort/steps/runtime: successful trials only

Case / variant	trials	Success	Control roughness overall (mean/transition)	Control roughness after wall (mean/transition)	Control effort	Number of steps	Average time per step (s)
condition=dynamic_wall variant=control_bank_mpp	400	296/400 (74.0%)	0.811 +/- 0.324 (n=296)	0.519 +/- 0.331 (n=296)	110.645 +/- 36.923 (n=296)	63.047 +/- 12.580 (n=296)	0.007 +/- 0.001 (n=296)
condition=dynamic_wall variant=corridor_prior_mpp	400	285/400 (71.2%)	0.772 +/- 0.240 (n=285)	0.605 +/- 0.291 (n=285)	117.579 +/- 34.818 (n=285)	63.302 +/- 14.245 (n=285)	0.005 +/- 0.001 (n=285)
condition=dynamic_wall variant=frenet_corridor_mpp	400	259/400 (64.8%)	0.561 +/- 0.235 (n=259)	0.468 +/- 0.273 (n=259)	107.726 +/- 32.643 (n=259)	65.807 +/- 13.563 (n=259)	0.007 +/- 0.001 (n=259)
condition=dynamic_wall variant=gaussian_prior_mpp	400	262/400 (65.5%)	0.548 +/- 0.227 (n=262)	0.476 +/- 0.261 (n=262)	107.566 +/- 34.657 (n=262)	65.737 +/- 14.006 (n=262)	0.008 +/- 0.001 (n=262)
condition=dynamic_wall variant=standard_mpp	400	42/400 (10.5%)	0.673 +/- 0.297 (n=42)	0.637 +/- 0.348 (n=42)	113.754 +/- 36.592 (n=42)	51.881 +/- 13.178 (n=42)	0.003 +/- 0.000 (n=42)
condition=no_wall variant=control_bank_mpp	100	96/100 (96.0%)	0.822 +/- 0.312 (n=96)	-	105.081 +/- 27.742 (n=96)	56.281 +/- 7.481 (n=96)	0.005 +/- 0.001 (n=96)
condition=no_wall variant=corridor_prior_mpp	100	92/100 (92.0%)	0.706 +/- 0.236 (n=92)	-	103.028 +/- 23.363 (n=92)	55.130 +/- 8.166 (n=92)	0.003 +/- 0.000 (n=92)
condition=no_wall variant=frenet_corridor_mpp	100	91/100 (91.0%)	0.565 +/- 0.221 (n=91)	-	103.984 +/- 24.115 (n=91)	59.330 +/- 7.497 (n=91)	0.005 +/- 0.001 (n=91)
condition=no_wall variant=gaussian_prior_mpp	100	96/100 (96.0%)	0.527 +/- 0.194 (n=96)	-	99.997 +/- 20.346 (n=96)	58.479 +/- 7.205 (n=96)	0.006 +/- 0.003 (n=96)
condition=no_wall variant=standard_mpp	100	35/100 (35.0%)	0.665 +/- 0.322 (n=35)	-	105.167 +/- 33.016 (n=35)	47.486 +/- 8.297 (n=35)	0.001 +/- 0.000 (n=35)
condition=static_wall variant=control_bank_mpp	400	392/400 (98.0%)	0.657 +/- 0.287 (n=392)	0.657 +/- 0.287 (n=392)	91.283 +/- 25.464 (n=392)	57.798 +/- 6.973 (n=392)	0.007 +/- 0.001 (n=392)
condition=static_wall variant=corridor_prior_mpp	400	388/400 (97.0%)	0.699 +/- 0.237 (n=388)	0.699 +/- 0.237 (n=388)	97.603 +/- 26.230 (n=388)	58.817 +/- 10.073 (n=388)	0.006 +/- 0.001 (n=388)
condition=static_wall variant=frenet_corridor_mpp	400	394/400 (98.5%)	0.476 +/- 0.226 (n=394)	0.476 +/- 0.226 (n=394)	93.659 +/- 28.053 (n=394)	59.726 +/- 7.727 (n=394)	0.007 +/- 0.001 (n=394)
condition=static_wall variant=gaussian_prior_mpp	400	382/400 (95.5%)	0.453 +/- 0.215 (n=382)	0.453 +/- 0.215 (n=382)	92.163 +/- 27.179 (n=382)	59.683 +/- 7.878 (n=382)	0.008 +/- 0.001 (n=382)
condition=static_wall variant=standard_mpp	400	58/400 (14.5%)	0.672 +/- 0.267 (n=58)	0.672 +/- 0.267 (n=58)	110.944 +/- 37.635 (n=58)	51.241 +/- 14.453 (n=58)	0.004 +/- 0.001 (n=58)

Failure reasons
Counts and percentages among failed trials

Case / variant	Failure reasons
condition=dynamic_wall variant=control_bank_mppi	collision: 104 (100.0%)
condition=dynamic_wall variant=corridor_prior_mppi	collision: 114 (99.1%); not_reaching: 1 (0.9%)
condition=dynamic_wall variant=frenet_corridor_mppi	collision: 138 (97.9%); not_reaching: 3 (2.1%)
condition=dynamic_wall variant=gaussian_prior_mppi	collision: 138 (100.0%)
condition=dynamic_wall variant=standard_mppi	collision: 349 (97.5%); not_reaching: 9 (2.5%)
condition=no_wall variant=control_bank_mppi	collision: 4 (100.0%)
condition=no_wall variant=corridor_prior_mppi	collision: 8 (100.0%)
condition=no_wall variant=frenet_corridor_mppi	collision: 9 (100.0%)
condition=no_wall variant=gaussian_prior_mppi	collision: 4 (100.0%)
condition=no_wall variant=standard_mppi	collision: 64 (98.5%); not_reaching: 1 (1.5%)
condition=static_wall variant=control_bank_mppi	collision: 8 (100.0%)
condition=static_wall variant=corridor_prior_mppi	collision: 8 (66.7%); not_reaching: 4 (33.3%)
condition=static_wall variant=frenet_corridor_mppi	collision: 6 (100.0%)
condition=static_wall variant=gaussian_prior_mppi	collision: 18 (100.0%)
condition=static_wall variant=standard_mppi	collision: 328 (95.9%); not_reaching: 14 (4.1%)